

Air Quality Management in California: Sensitivity to Changing Climate



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Acknowledgments

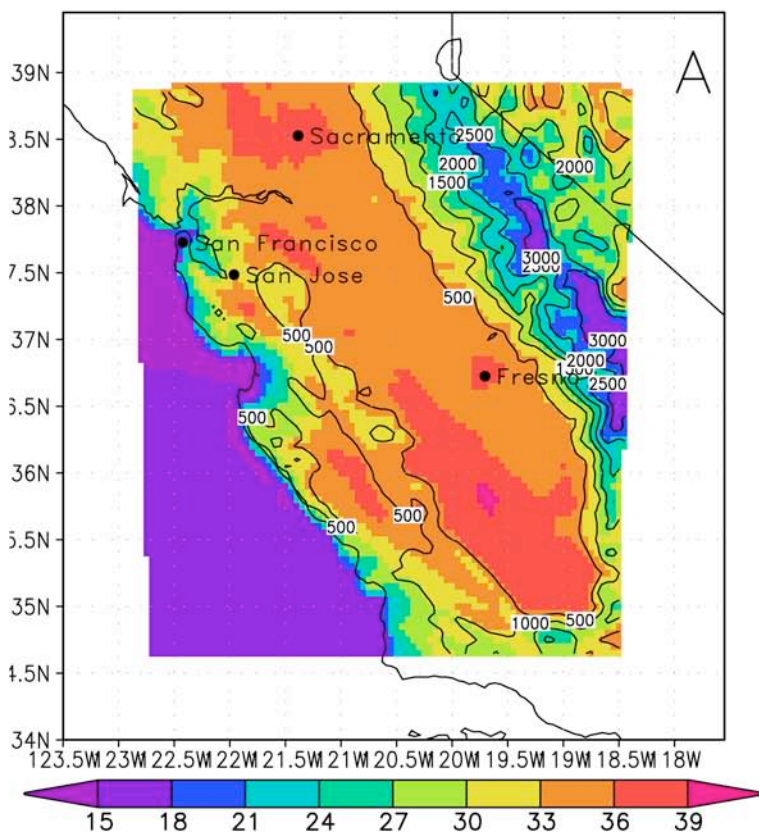
- UC Berkeley:
 - » Ron Cohen, Allen Goldstein, Allison Steiner, Shaheen Tonse
- Technical Assistance:
 - » BAAQMD, CARB, LBNL, NOAA
- Financial Support:
 - » U.S. EPA

Air Quality Modeling

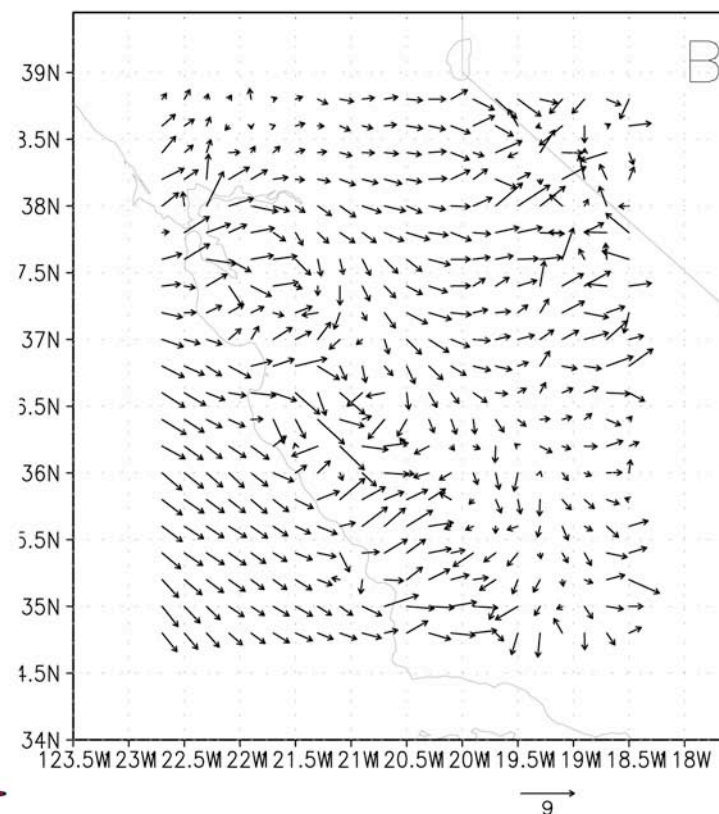
- Use CMAQ model with SAPRC99 chemistry to predict ozone in Central California
- Base case episode from summer 2000
- Consider effects of changes in:
 - » Reaction rates (increased T)
 - » Biogenic emissions (increased T)
 - » Anthropogenic emissions (population growth and technology change)
 - » Inflow boundary conditions (global change)

Modeling Domain

Temperature & Topography



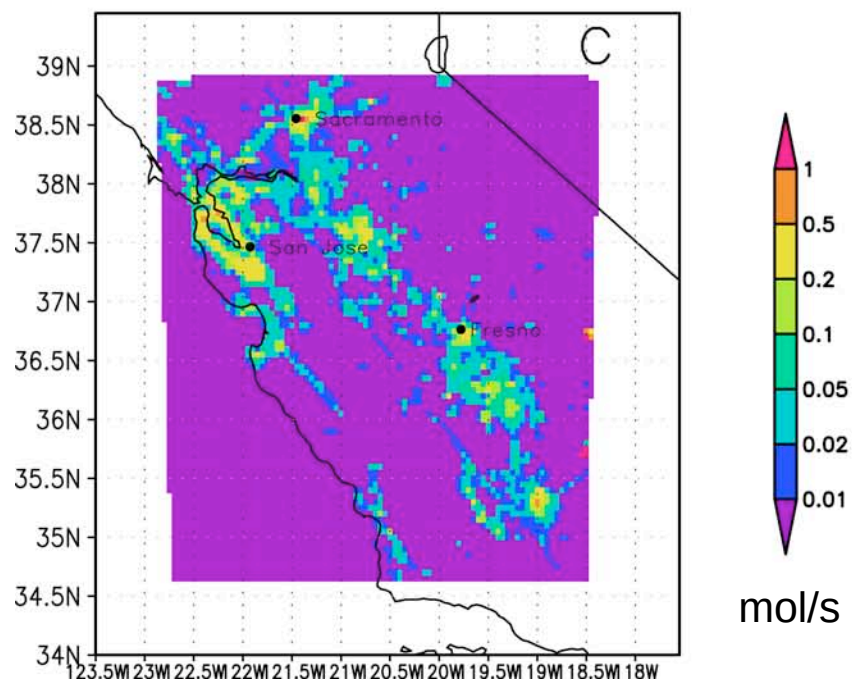
Winds



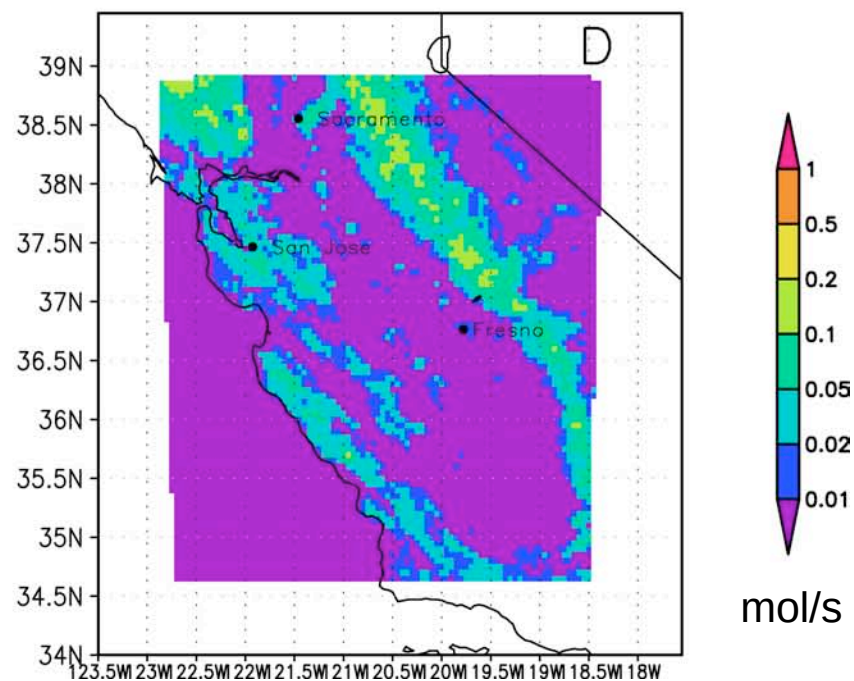
MM5 (Wilczak et al NOAA) for July 29-Aug 2, 2000

VOC Emissions

Anthropogenic



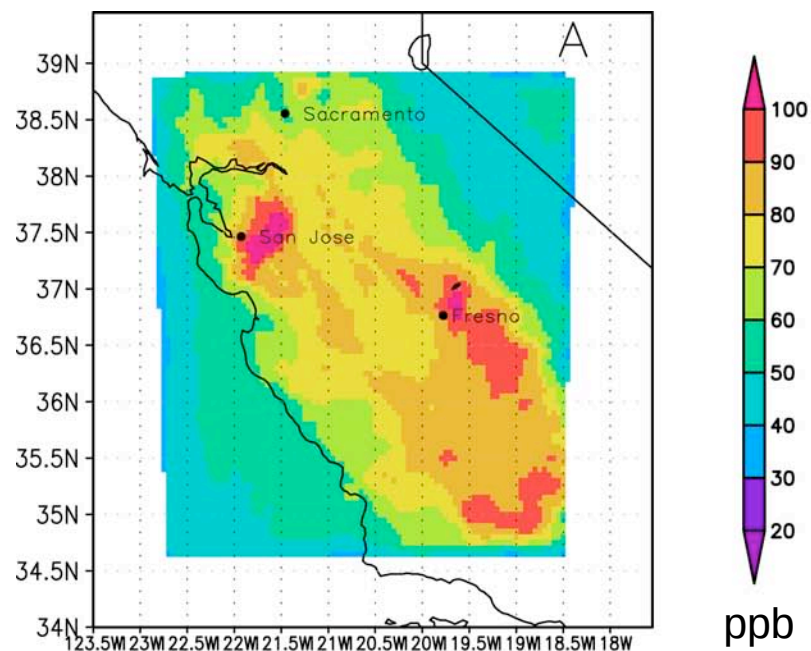
Biogenic



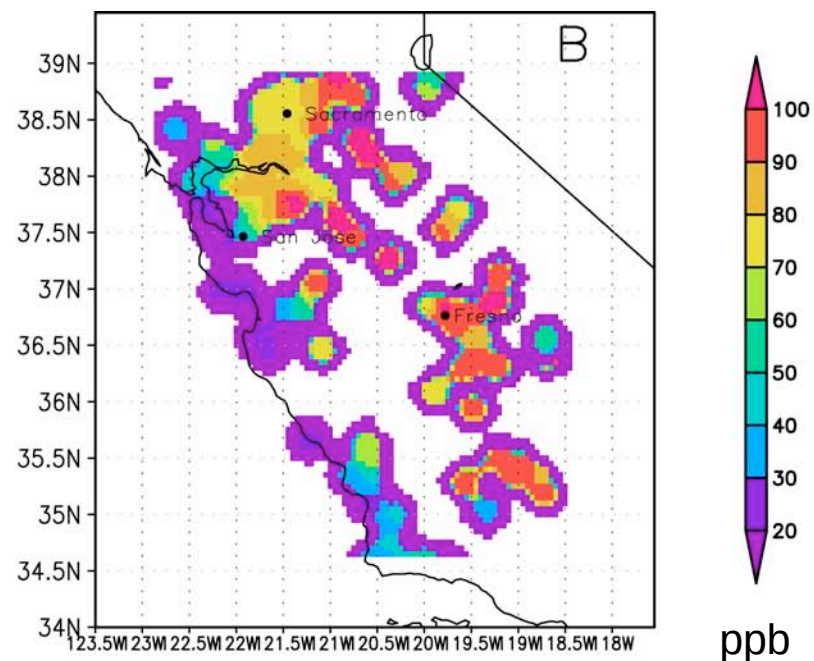
Emission rates shown for 3 PM

Base Case Results

Predicted O_3



Observed O_3



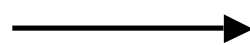
Ozone at 3 PM for 3rd-5th days of Jul 29-Aug 2 episode

Climate Change

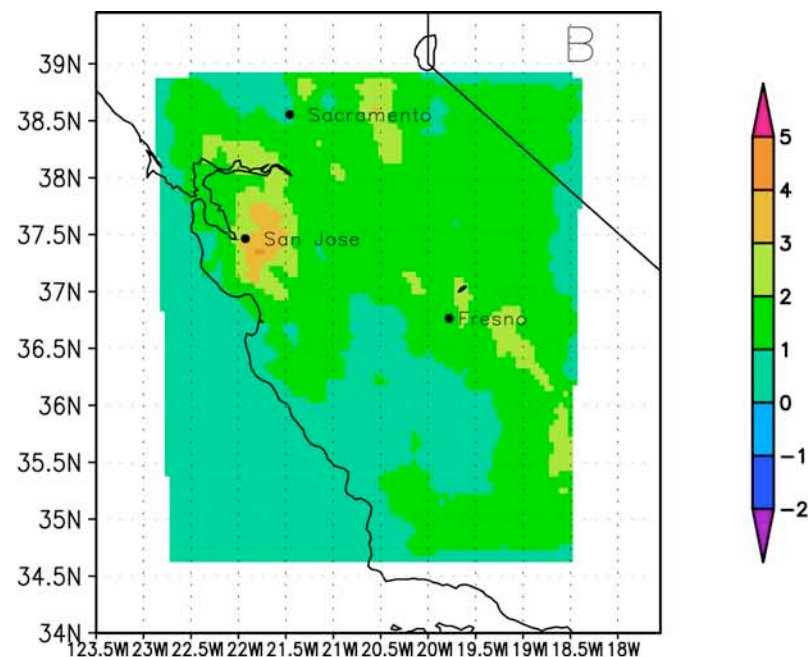
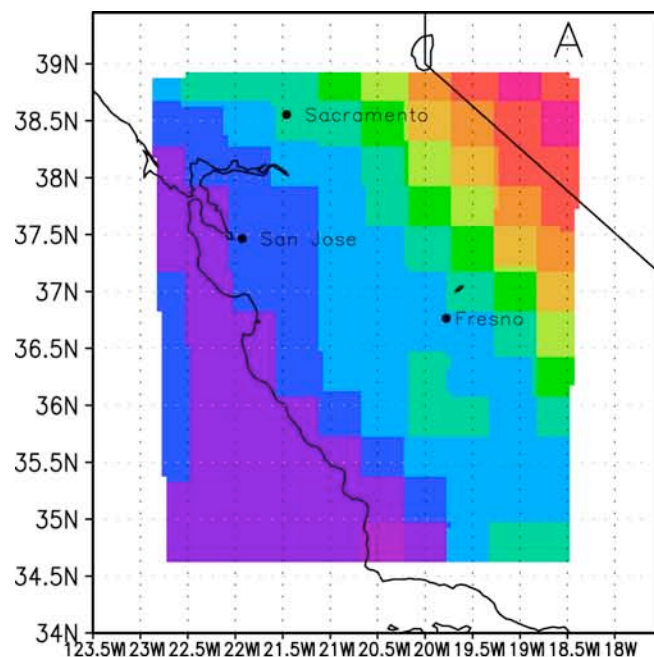
- Consider effects of CO₂ doubling
 - » Pre-industrial = 280 ppm
 - » Future year (~2050) = 560 ppm
- Snyder et al. (*GRL* 2002) used community climate model (CCM3) to drive regional climate simulations for California at 40 km resolution

Effect of ΔT on Chemistry

Change in T ($^{\circ}\text{C}$)



Change in O_3 (ppb)



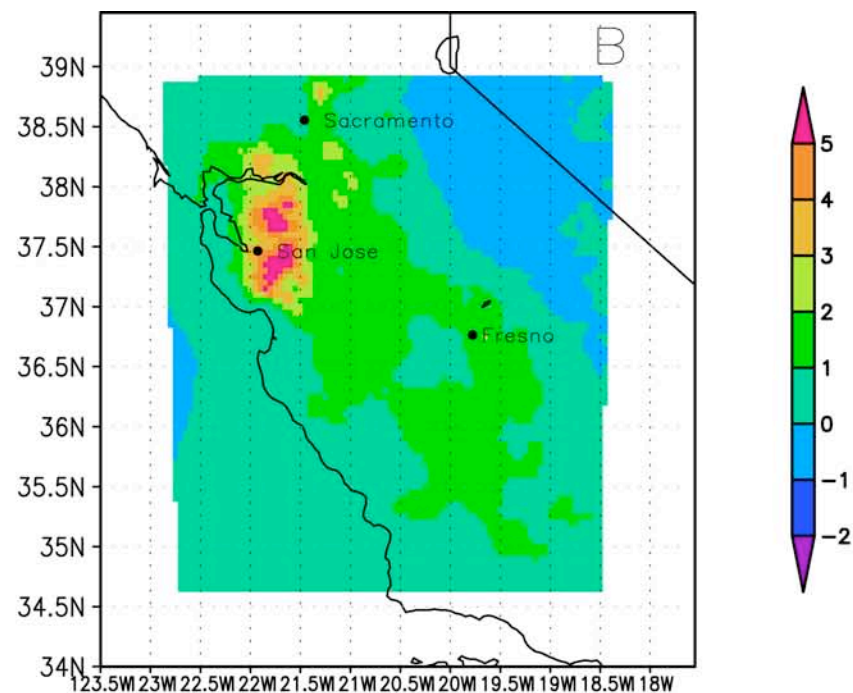
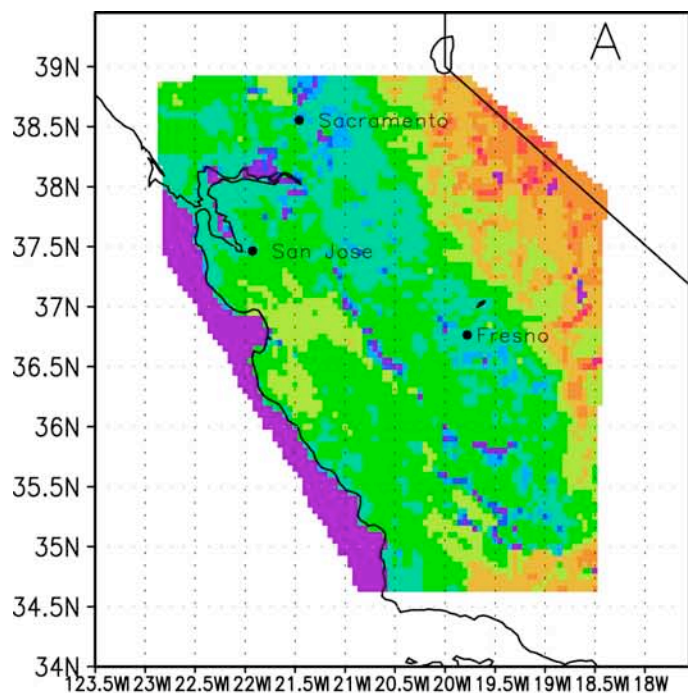
ΔT for month of August from Snyder et al. (2002)
Other meteorological variables & emissions unchanged

Effect of Δ BVOC on Ozone

% Increase in BVOC



Change in O₃ (ppb)

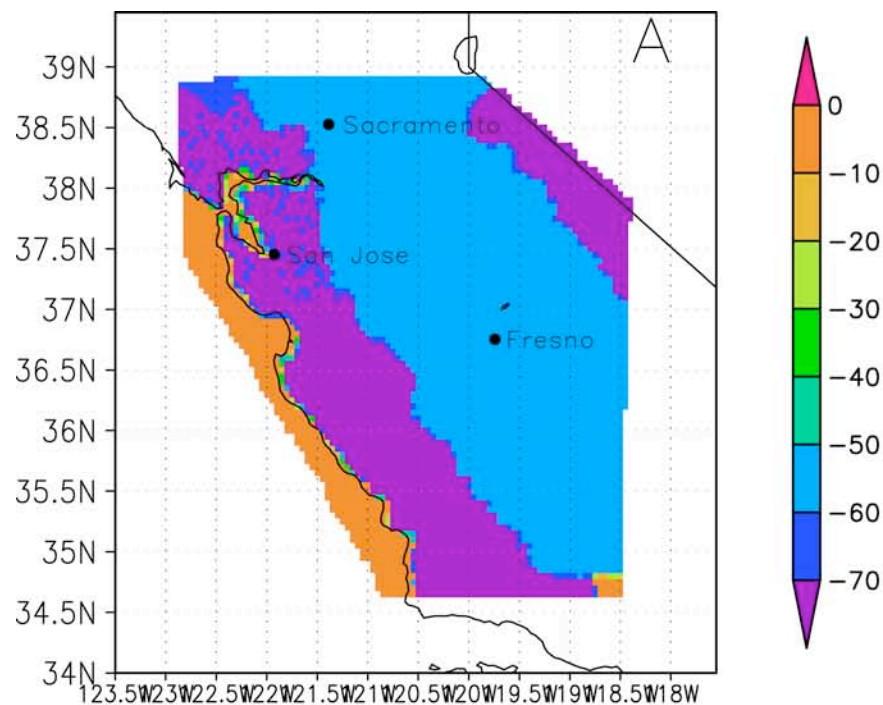


Biogenic emissions of isoprene & MBO peak at 37°C

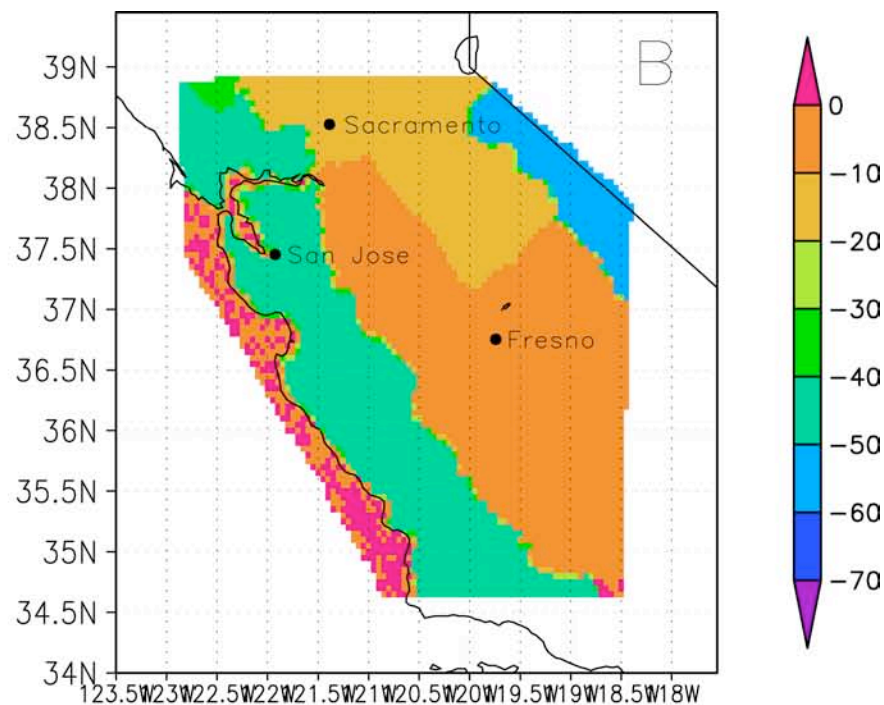
Terpene emissions increase exponentially with T

Anthropogenic Emissions: 2050

% Change in VOC & CO



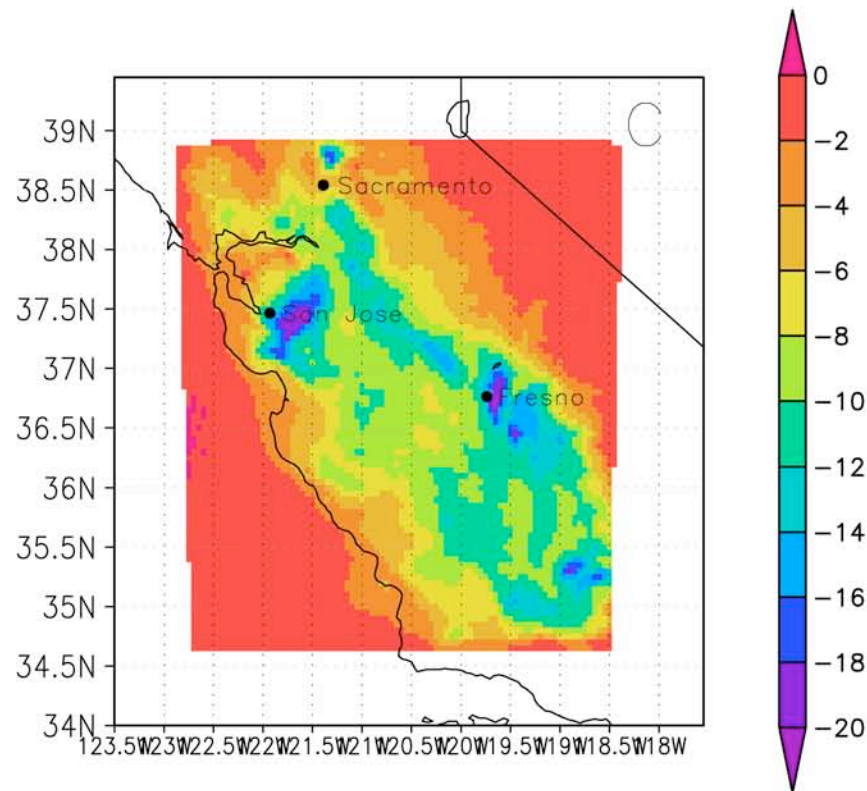
% Change in NO_x



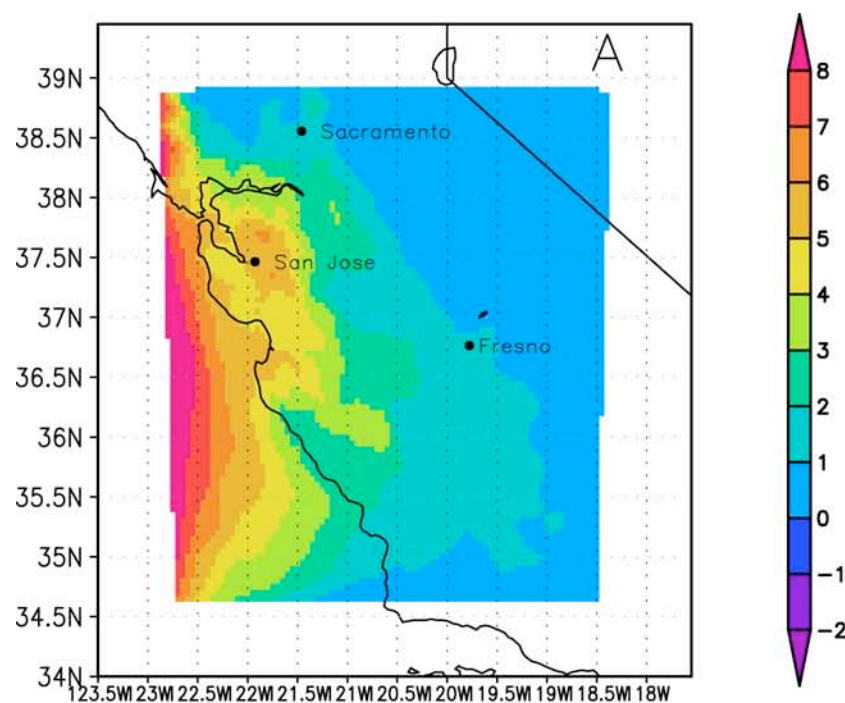
Population forecasts: faster growth in Central Valley
Growth factor for NO_x is 2× that for VOC & CO
80% reduction in present-day emission factors for
CO, VOC & NO_x

Future Emissions & Inflow BCs

ΔO_3 (ppb) with
2050 Emissions



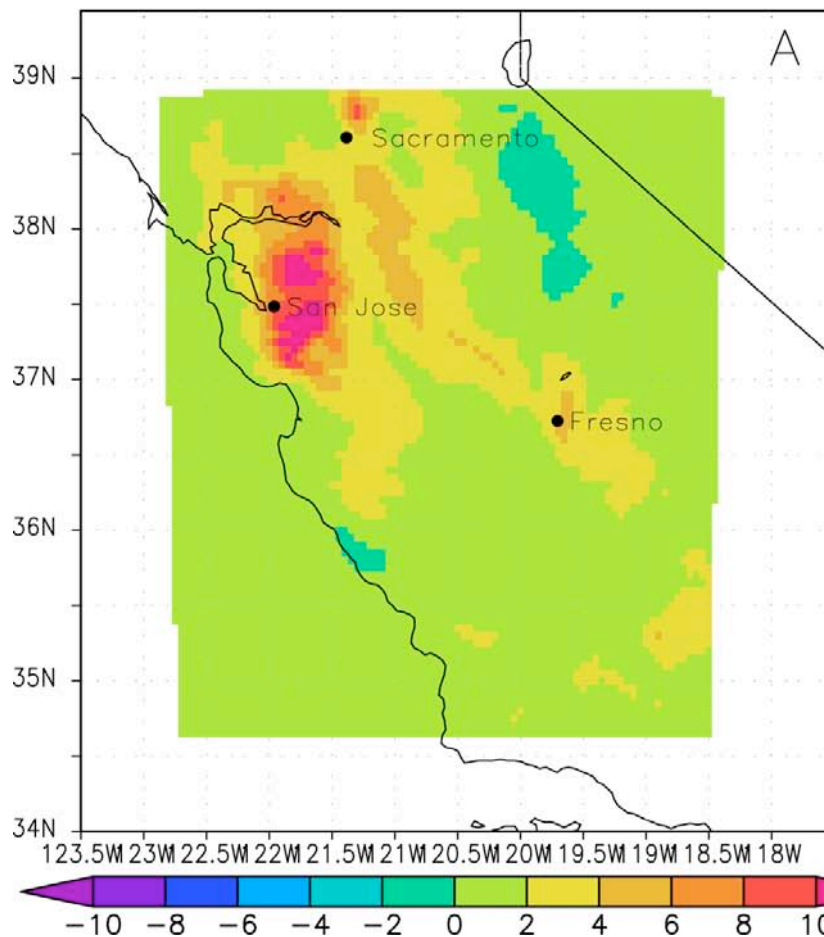
ΔO_3 (ppb) with
2050 Inflow BC



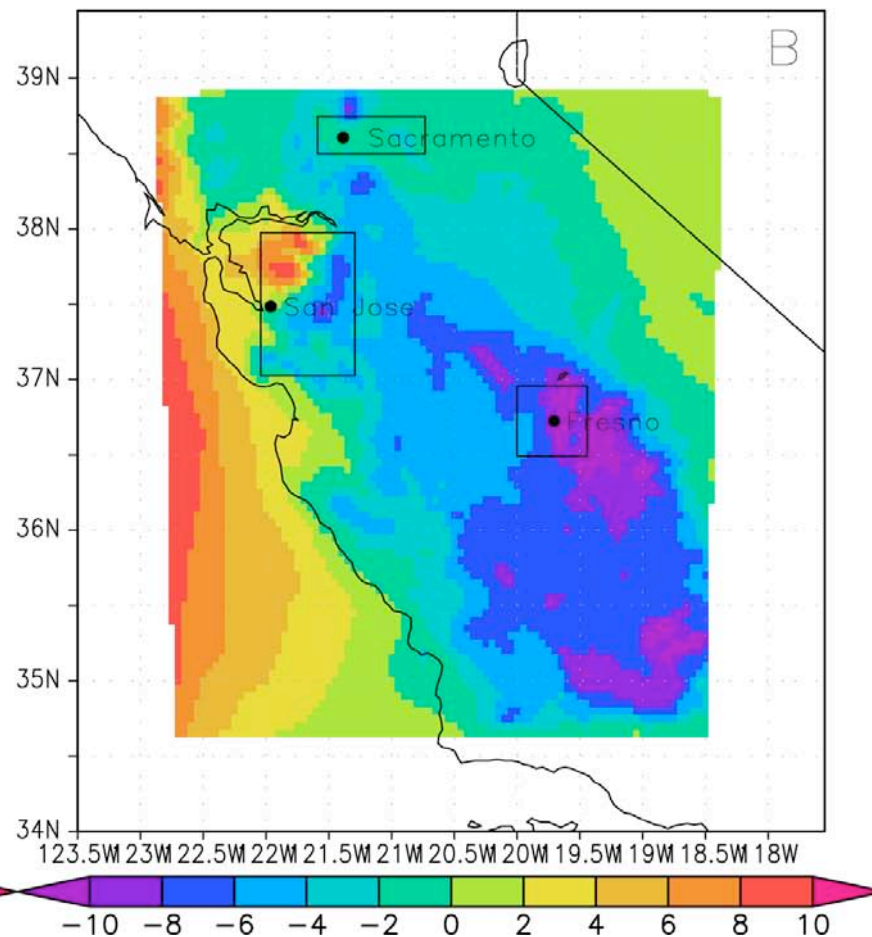
CO: 80 → 104 ppb
CH₄: 1.7 → 2.4 ppm
O₃: 30 → 40 ppb

Combined Simulations

Temperature Effects



ΔT + 2050 Emissions/BCs

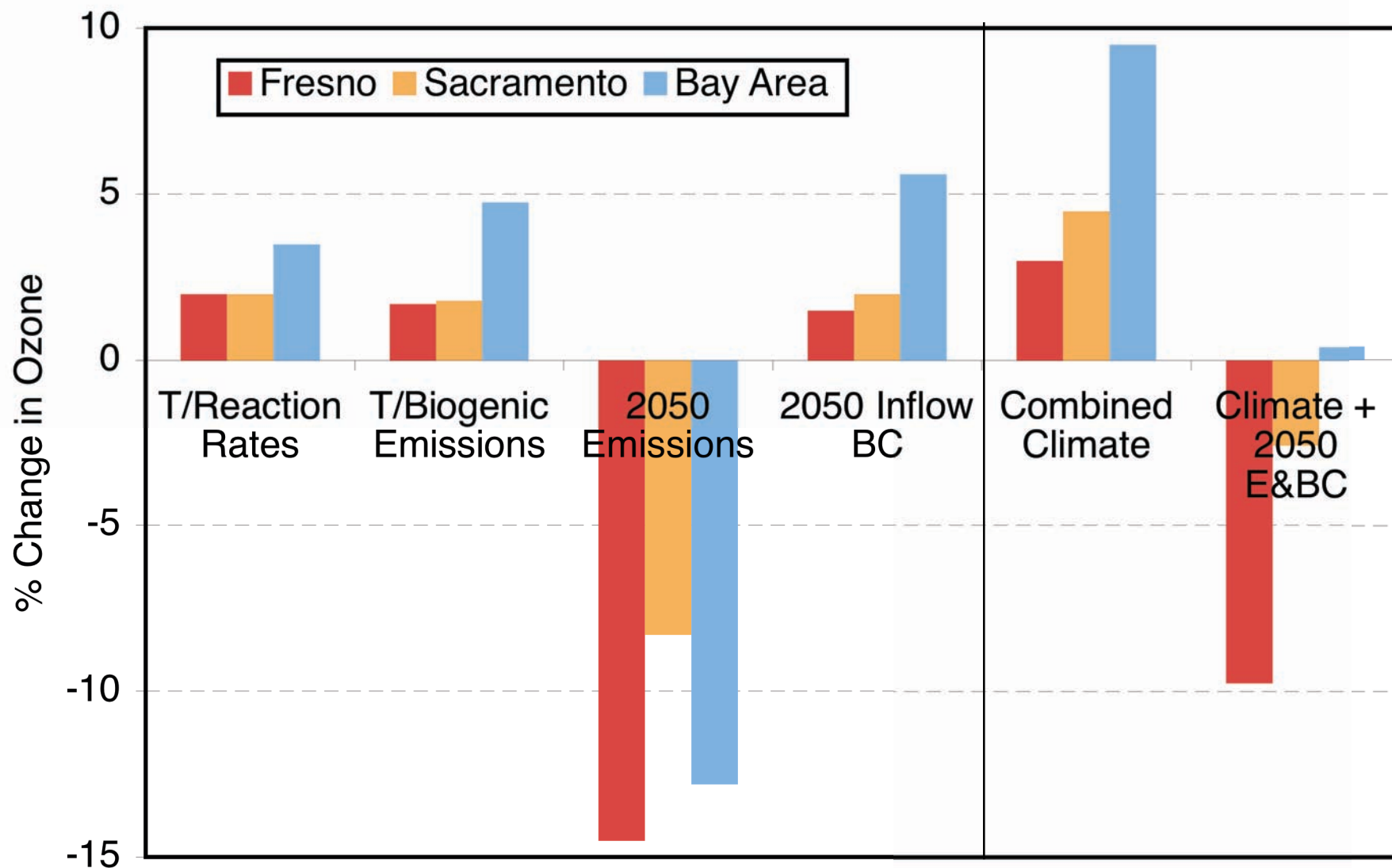


ΔO_3 (ppb)

Other Effects

- Effects on water resources (see Snyder et al.)
 - » rainfall increases up to 20% in northern California
 - » snowpack decreases ~80% by end of Feb
- Sea level rise
- Changes in forest fires
- Changes in PM_{2.5} & winter season air quality
- Ecosystem and land use change
- Health effects of extreme hot weather
- Changes in frequency of O₃-conducive meteorology & length of high-O₃ season

Summary of O₃ Effects



Key References

- Snyder et al. (2002). Climate responses to a doubling of atmospheric CO₂ for a climatically vulnerable region. Appears in *Geophysical Research Letters (GRL)*.
- Steiner et al. (2006). Influence of future climate and emissions on regional air quality in California. Appears in *Journal of Geophysical Research (JGR)*.